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# Adaptive Regulation of Micro Closed-Loop Space Station Ecosystems Using Proximal Policy Optimization

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## Abstract

Long-term orbital missions require highly independent life-support systems, maintaining ecological stability under limited external support and subsystem attenuation. Conventional strategies are typically based on fixed threshold rules or deterministic mass-balance models and cannot adapt to disturbances or multi-objective trade-offs between resource safety and environmental load. This paper proposes a reinforcement learning framework for managing micro closed-loop ecological systems under reliability decay and environmental constraints. A reinforcement learning controller based on Proximal Policy Optimization (PPO) is trained to learn stable regulation policies with random disturbances. A dynamic state model is constructed to describe the coupled and time-sequential elements, ranging from water storage, oxygen concentration, energy reserve, subsystem reliability to ecological load. Degradation propagation is integrated into the transition dynamics through reliability-dependent recycling efficiency. Simulation experiments demonstrate that the proposed policy maintains all resource variables within the safety margins while reducing ecological imbalance compared with traditional strategies. Consequently, the proposed framework provides an expandable control architecture for independent regulation of space station life-support systems.

## Keywords

Proximal Policy Optimization (PPO); Micro closed-loop space station ecosystems; Sensitivity analysis

## Introduction

Long-term space station missions require reliable ecological management systems to maintain resource cycles under strict constraints, ranging from mass, energy to reliability [2],[4],[5],[6],[7]. Different from short-term missions, long-term space station operations cannot rely on frequent external support, but only depend on internal micro closed-loop life-support systems to secure the environmental stability. Disturbance in one subsystem may propagate through resource flows, which may cause a series of impacts across the overall ecological system. Therefore, maintaining stable control of integrated cycles is a critical challenge for long-term missions [9].

Traditional life-support controlling strategies are typically based on fixed threshold control [3],[5]. These methods assume predictable subsystem performance. They only respond when system variables approach predefined safety limits. However, subsystem attenuation, random disturbances, and environmental fluctuations become increasingly significant during long-term operations [5]. Therefore, deterministic control strategy model cannot effectively react to nonlinear or urgent issues between dynamics and reliability. Consequently, rule-based and fixed strategies may lead to inefficient resource allocation or delayed responses, causing ecological imbalance.

Recent advances in reinforcement learning have provided effective tools for independent decision-making in complex systems. In this study, a reinforcement learning framework for micro closed-loop space station ecosystems is developed using Proximal Policy Optimization (PPO). A compact state-space model is constructed to represent the integrated elements of water storage, oxygen concentration, energy reserve, subsystem reliability, and ecological load. The proposed model enables adaptive regulation policies, maintaining ecological stability under disturbances and subsystem degradation. In addition to regulation performance, this study further examines policy response characteristics and discusses deployment-oriented computational considerations for autonomous onboard control.

The main contributions of our study are multi-dimensional. First of all, a ecological dynamics model is developed to combine resource balance, reliability, and environmental load. Secondly, the life-support problem is formulated as a sequential decision process with random degradation and integrated ecological transitions. Third, a PPO-based controller is introduced in this paper and shown to improve smoothness and long-lasting ecological performance.

## Micro Closed-Loop Ecosystem Modeling

### *State Representation*

The micro closed-loop life-support system is represented by a compact state vector, which captures the most essential resource, reliability, and environmental elements. At decision step $t$, the system state is defined as

$$x_t = [W_t, O_t, E_t, R_t, L_t]^T \quad (1)$$

$W_t$ denotes the normalized water storage level, $O_t$ represents the normalized oxygen concentration, $E_t$ is regarded as the available energy reserve, $R_t$ denotes the overall reliability of the integrated subsystems, and $L_t$ is the ecological load index associated with internal environmental stress. These five parameters were selected because they represent short-term availability, and long-term subsystem health. However, although the ecological system in our real life contains more physical quantities, our representation reserves the most eseential information needed for regulation.

### *Water Resource Dynamics*

Water balance is determined by crew consumption, recovery, and external supply [5],[8]. $P$ represents the crew population size, $\omega$ is the average water consumption rate, $\eta_w(R_t)$ is the recycling efficiency with the current subsystem reliability level, and $u_{w,t}$ denotes the supplemental water supply applied at step $t$. The corresponding sequential state can be expressed as

$$W_{t+1} = W_t - \omega P + \eta_w(R_t)\,\omega P + u_{w,t} \quad (2)$$

The second term represents net water demand from crew activity. And the third term reflects the proportion of consumed water recovered from the recycling subsystem. The last term is introduced as a controllable input, which compensates for temporary imbalance when recycling performance deteriorates.

### *Oxygen Balance*

Oxygen evolution is modeled in a similar way as the water, considering metabolic consumption and regeneration from life-support system. In oxygen update is written as

$$O_{t+1} = O_t - \rho P + \eta_o(R_t)\,\rho P \quad (3)$$

where $\rho$ is the average oxygen consumption rate, which is based on the previous statistic and $\eta_o(R_t)$ represents the oxygen regeneration efficiency under the current reliability state. This equation captures the fact that oxygen recovery capacity is also affected by subsystem attenuation. When reliability decreases, the effective regeneration rate declines, increasing the risk of ecological imbalance.

### *Energy Balance*

Energy is treated as a shared supporting resource required for recycling and regulation processes. $c_w$ and $c_r$ represents the energy coefficients related to water supply and recycling regulation, respectively. $u_{r,t}$ is considered as the recycling control intensity, and $g_t$ represents the available energy input at step $t$. The energy reserve evolves according to

$$E_{t+1} = E_t - c_w u_{w,t} - c_r u_{r,t} + g_t \quad (4)$$

Stabilizing the system requires energy expenditure, and intense control actions may reduce operational flexibility in the future when the energy storage is insufficient.

### *Reliability Modeling*

Long-term operation undoubtedly leads to subsystem attenuation [5],[8]. To capture it, the overall reliability of the ecological system can be modeled as

$$R_t = \sum_{i=1}^{N} \omega_i r_{i,t}, \sum_{i=1}^{N} \omega_i = 1 \quad (5)$$

where $(r_{i,t})$ is the reliability of subsystem $(i)$ at decision step $(t)$, $(\omega_i)$ is its weighting coefficient, and $(R_t)$ represents the overall reliability of the life-support subsystems.

$$R_{t+1} = R_t - \alpha R_t + \epsilon_t \quad (6)$$

In which $\alpha$ denotes the baseline attenuation rate and $\epsilon_t$ represents random disturbances, including minor recovery effects, temporary fluctuations, or rare shocks. This equation provides a simplified approximation of cumulative subsystem degradation.

To reflect the influence of reliability on recycling performance, the water recovery efficiency is assumed to satisfy

$$\eta_w(R_t) = \eta_{w0} R_t \quad (7)$$

where $\eta_{w0}$ is the nominal recycling efficiency in ideal conditions. This relation means that degradation in subsystem reliability directly reduces recovery effectiveness, thereby integrating long-term equipment condition with short-term resource stability.

### *Ecological Load Evolution*

In addition to direct resource balance, the internal environmental burden of the habitat must also be considered. Ecological load is used here as an accumulated indicator of environmental stress caused by persistent resource and operational inefficiency. Its evolution can be modeled as

$$L_{t+1} = L_t + \beta_w(W_t - W^*)^2 + \beta_o(O_t - O^*)^2 + \beta_e(E_t - E^*)^2 \quad (8)$$

$W^*$, $O^*$, and $E^*$ represents the nominal target values of water, oxygen, and energy, respectively, and $\beta_w$ , $\beta_o$ , and $\beta_e$ are nonnegative weighting coefficients. This expression represents the persistent difference from ideal operating conditions and allows long-term imbalance to accumulate gradually over time. This accumulated form is important because risks in long-term habitats do not arise from a single deviation. It always comes from persistent wrong operation. In that case, the load variable acts as a slow indicator that complements the resource states.

## Reinforcement Learning Formulation for Ecological Regulation

The life-support regulation problem is constructed as a reinforcement learning task for sequential decision-making under uncertainty[2]. At each decision step, the monitor observes the current state and determines how much regulation effort should be allocated to water support, energy redistribution, and recycling adjustment.

### *State Space*

Specifically, the state at step $t$ is defined as

$$s_t = x_t = [W_t, O_t, E_t, R_t, L_t]^T \quad (9)$$

### *Action Space*

The regulation action is represented by a continuous control vector

$$a_t = [u_{w,t}, u_{e,t}, u_{r,t}]^T \quad (10)$$

### *Transition Dynamics*

Given the current state $s_t$ and control action $a_t$, the ecological system evolves according to the coupled nonlinear dynamics conditions defined previously. The corresponding transition equation can be written in compact form as

$$s_{t+1} = f(s_t, a_t) + \xi_t \quad (11)$$

In which $f(\cdot)$ represents the deterministic component of the ecological dynamics and $\xi_t$ denotes random rare disturbances.

### *Safety Constraints*

To ensure safe operation of the micro-closed loop ecosystem, the regulation process is subject to hard ecological constraints. These constraints are imposed on the primary resource variables as

$$\begin{aligned} W_t &\geq W_{\min} \\ O_{\min} &\leq O_t \leq O_{\max} \\ E_t &\geq E_{\min} \end{aligned} \quad (12)$$

$W_{\min}$, $O_{\min}$, $O_{\max}$, and $E_{\min}$ represents the minimum safety bounds. Violation of these conditions indicates that the ecological system has entered an unsafe and unstable operating region.

### *Reward Function*

The reward function is designed to represent both ecological imbalance and unsafe control behavior. It is regarded as

$$r_t = -\lambda_1 L_t - \lambda_2 \Phi(s_t, a_t) \quad (13)$$

The coefficient $\lambda_1$ controls the relative importance of long-term ecological stability, while $\lambda_2$ determines the severity of penalties imposed on unsafe or inefficient control behavior. The reward in this function is designed to balance two competing objectives. On the one hand, the controller should keep the variables close to their normal operating conditions. On the other hand, it should avoid excessive effort or unsafe regulation, which may stabilize one resource while increasing long-term burden.

## Policy Optimization Using Proximal Policy Optimization

To solve the sequential ecological regulation problem defined in the previous section, a reinforcement learning method based on PPO is adopted. In contrast, PPO is well suited for this setting because it supports continuous action spaces and provides stable policy updates through a clipped objective.

Let $(\pi_\theta(a_t|s_t))$ denote the PPO policy and let $(\widehat{A_t})$ denote the estimated advantage function at decision step $(t)$[1]. The PPO objective is written as

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(r_t(\theta)\widehat{A_t},\ \text{clip}(r_t(\theta),\ 1-\epsilon,\ 1+\epsilon)\widehat{A_t}\right)\right] \quad (14)$$

In which $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio between the updated policy and the previous policy, and $\epsilon$ is the clipping coefficient that limits the magnitude of policy change during each update. By constraining the update step in this way, PPO reduces the risk of destructive parameter shifts and improves training stability under uncertain transitions.

The reinforcement learning controller consists of a policy network and a value network. Rectified Linear Unit activation is used in the hidden layers, and the policy is optimized using the Adam optimizer. Generalized Advantage Estimation is employed to reduce variance in the gradient update [1]. During training, the initial ecological states are randomly sampled within admissible operating ranges, and each episode evolves according to the stochastic transition model defined in previous section . At every decision step, the policy outputs sequential actions for accumulated water support, energy allocation, and recycling intensity. The resulting trajectories are then used to evaluate returns and update the policy iteratively until convergence. The training process converges steadily, as illustrated in Figure 1.

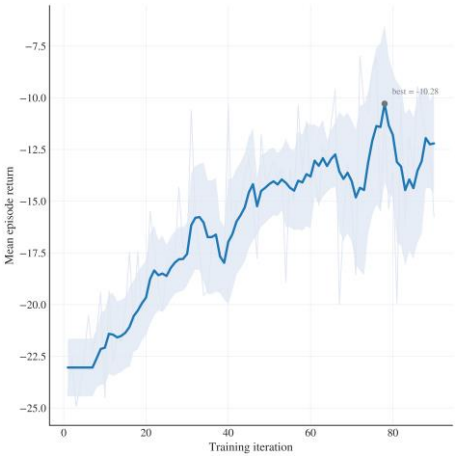



Figure 1. Training reward convergence during PPO optimization.

As shown in Figure 1, the average training reward increases steadily during optimization and gradually approaches a stable number, indicating that the controller learns an effective regulation strategy under the random transition model. By the way, the variance of episodic return decreases over training, suggesting improved policy consistency and reduced sensitivity.

A practical configuration is to use a learning rate of $3 \times 10^{-4}$, a discount factor of 0.99, a clipping coefficient of 0.2, and a mini-batch size of 64. The maximum number of training iterations may be set to $1.0 \times 10^{5}$ or higher depending on convergence behavior.

This choice is constructed by three considerations below. First of all, the regulation problem involves continuous actions rather than discrete decisions. Secondly, the clipped PPO objective helps to prevent destructive policy updates. Thirdly, the method has shown good empirical stability in uncertain scenarios. This is especially relevant here because the controller must balance immediate correction against long-term sustainability. A limited strategy may reduce local fluctuations in the short term, but it cannot easily coordinate the delayed interaction between multi-dimensions.

## Simulation and Results

### *Experimental Setup*

Simulation experiments were conducted over a horizon of 500 decision steps in a bid to evaluate the long-term regulation performance of the proposed method under random degradation. At the beginning of each episode, the state variables were initialized within admissible operating ranges around the nominal equilibrium point, so that the system started from a feasible but not perfectly balanced ecological condition. During simulation, subsystem reliability decayed progressively according to the degradation model introduced in previous section, while random disturbances were injected into the transition dynamics to represent uncertainty in recovery efficiency, energy availability, and environmental fluctuation.

The attenuation rate was to represent a long-term attenuation scenario. The environmental weighting coefficients were tuned so that ecological load accumulated gradually under persistent deviation It do not dominate the short-term dynamics. This setting allows the simulation to distinguish between short-term controlling quality and long-term burden.

The PPO policy was trained using the parameter settings reported in the previous section. The baseline controller was implemented using fixed safety thresholds and rule-based corrective actions, which provides a natural reference because such logic is widely used in practical ecological regulation. This baseline was not intended to represent the best possible controller, but rather a practical reference with transparent logic and clear safety-oriented behavior. The comparison highlights whether adaptive learning adds value beyond ctraditional threshold-triggered correction or not. All simulations and PPO training were implemented in Python 3.10 with PyTorch 2.2 and CUDA 12.1 on an NVIDIA RTX 4090 GPU. Each experiment was repeated over 50 independent runs with different random seeds, and the reported results are averages over these runs. The present study is a simulation-based proof of concept rather than a flight-qualified validation campaign. No operational telemetry from closed-loop life-support platforms was available in the current stage, and no hardware-in-the-loop testbed was incorporated. Therefore, the reported results verify the control feasibility of the proposed PPO framework under structured stochastic disturbances, but do not yet establish full engineering readiness. A practical validation pathway would include disturbance replay using telemetry-like subsystem traces, followed by hardware-in-the-loop evaluation of policy latency, memory footprint, and closed-loop stability on resource-constrained onboard processors.



### *Resource Stability*

The PPO-based policy maintained all major resource variables within admissible operating ranges throughout the simulation horizon. Compared with the threshold-based controller, the PPO-based policy exhibited smaller fluctuation and smoother recovery behavior after disturbances. As shown in Figure 2, the PPO-based controller maintains smoother water, oxygen, and energy trajectories than the threshold-based baseline. This difference is especially evident in the water and oxygen trajectories, where the baseline controller tended to react only after the state approached a safety boundary, whereas the learned policy adjusted regulation intensity earlier and avoided large excursions from the nominal operating point. The proposed policy reduced water storage variation by approximately 42%, oxygen concentration fluctuations were moderately reduced, and energy variance by 25% relative to the threshold-based strategy.

More importantly, all primary resource parameters remained within accepteable safety edges across the random runs, indicating that the smoother trajectories were not achieved at the cost of reduced safety. The main benefit of the learned policy lies in reducing peak fluctuations. It also avoids delayed corrective actions near the boundary.

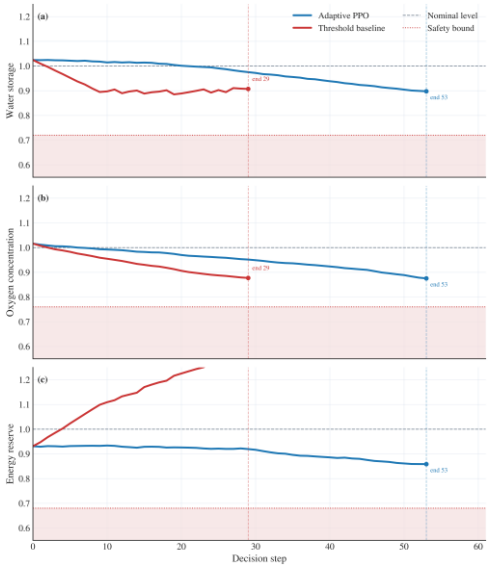



Figure 2. Resource state trajectories under adaptive policy and threshold control.

### *Ecological Load Suppression*

Figure 3 shows that the ecological load grows more slowly under the PPO-based policy than under the threshold-based controller.

Under the PPO-based reinforcement learning policy, ecological load accumulation was reduced by approximately 38% compared with the threshold-based controller. This improvement indicates that the PPO policy was able to coordinate resource recovery and energy

expenditure more effectively over time, instead of relying on short-term corrective actions. In particular, the learned controller tended to suppress deviation earlier, which limited the cumulative growth of environmental burden across the episode.

This result is important because ecological load reflects not only instantaneous departure, but also the long-term accumulation of internal environmental pressure. In that case, a lower growing rate of environmental load suggests that the controller improves sustainability over extended operation rather than merely correcting short-term imbalances.

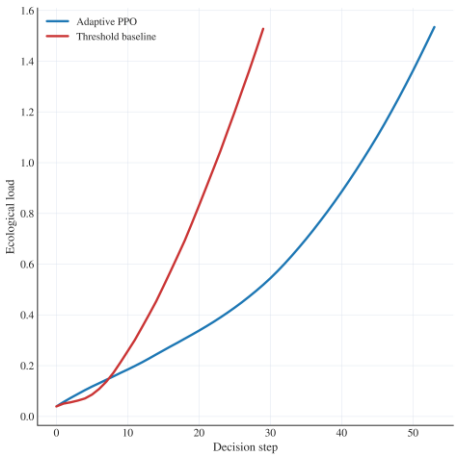



Figure 3. Ecological load evolution under adaptive PPO regulation and threshold-based control.

### *Robustness Under Degradation*

To evaluate robustness, the degradation coefficient was varied across multiple scenarios while keeping the control structure unchanged. This experiment was designed to test whether the learned policy remained effective when the system worsens faster than the ideal case. The results show that the PPO-based adaptive policy preserved stable operation over a wider degradation range than the threshold-based baseline. This trend is evident in Figure 4, which compares system performance across multiple degradation scenarios. Even when degradation was accelerated, the learned policy continued to allocate regulation effort in a balanced manner and prevented abrupt ecological collapse.

This experiment can also be interpreted as a one-dimension sensitivity analysis with respect to reliability attenuation. By disturbing the attenuation coefficient and keeping the policy structure fixed, we evaluate whether the learned controller preserves acceptable quality under parametric model shift rather than only under a single scenario.

An important result is that the proposed strategy extended safe operational duration by nearly 90% under accelerated degradation. Equally important, the policy does not fail accidentally when the degradation rate increases. Instead, its performance worsen more gradually, which is a acceptable property for random environmental control because it preserves a wider intervention window before the habitat approaches unsafe region.

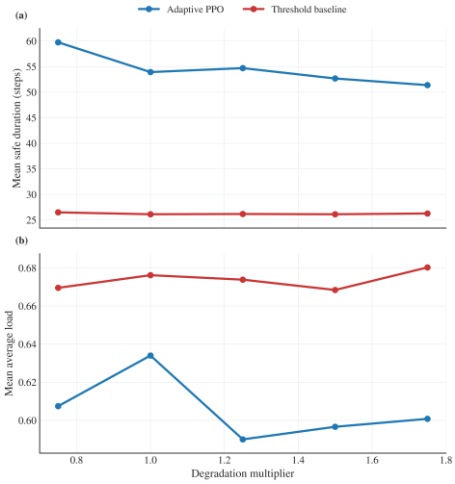



Figure 4. System performance under different subsystem degradation rates.

### *Policy Response Analysis*

To clarify the decision-making logic of the learned controller, we examined how the policy responds to local perturbations in individual state variables while holding the remaining states fixed. The analysis shows that the supplemental water action increases sharply as the water state approaches its lower safety bound, whereas recycling intensity becomes more aggressive when subsystem reliability declines. In contrast, the policy suppresses energy-intensive regulation when the energy reserve is already limited. This behavior indicates that the policy implicitly learns a prioritization structure across integrated environmental risks. Instead of maximizing a single resource variable, it reallocates controlling effort. It is according to which subsystem is closest to instability and which intervention is energetically affordable at that moment.

From an engineering perspective, this response pattern suggests that the controller is not following a single fixed order. Instead, it adapts its effective priority according to the evolving joint state of resources, reliability, and energy availability. Such behavior is desirable in long-duration habitats. Safe controlling depends less on fixed threshold and more on balancing multiple risks before any single parameter reaches an unrecoverable region.

## Discussion

Compared with threshold-based control, the learned policy exhibits earlier adjustment and lower cumulative environmental burden in the tested stochastic scenarios. In this sense, the learned controller behaves as a risk-aware allocator rather. It tends to redistribute controlling effort before large disparity emerge. It helps explain why envrionmental load increases more tenderly under our proposed adaptive policy.

Another important implication of the results is that subsystem reliability should not be treated as an external background factor. In the framework, reliability is explicitly coupled with recycling efficiency and ecological load evolution. This integrating implies that long-term equipment attentuation should be treated as part of the control state rather than as an single external issue. In engineering perspective, envrionmental controlling and subsystem robustness management become partially intertwined over long-term mission durations.

The same control architecture can be extended to additional modules such as waste processing, thermal regulation, or carbon dioxide control.

Offline training should be distinguished from onboard execution [10]. In this study, PPO training used an RTX 4090 GPU, but onboard deployment would require policy inference only rather than policy optimization. Because the state and action spaces are compact, the inference burden is expected to be much lighter than the training burden. Practical deployment would still require explicit benchmarking of latency, memory footprint, and power consumption on target avionics platforms, and periodic offline retraining is more realistic than continuous onboard updating.

At the same time, several limitations should be acknowledged. The current state-transition model is intentionally simplified in a bid to make policy learning tractable within a compact framework. Some physical processes, ranging from detailed subsystem latency, maintenance scheduling to nonlinear cross-coupling between environmental variables, are not explicitly modeled. In addition, the present validation remains limited to synthetic disturbance scenarios and simulated subsystem degradation. As a result, the reported gains should be interpreted as evidence of control potential under a structured stochastic model, rather than as a direct guarantee of performance in operational space habitats. Stronger engineering claims would require validation against telemetry-derived disturbance profiles, subsystem-specific degradation data, and hardware-in-the-loop execution tests. Even so, the simplified model remains useful as an intermediate research step. It isolates the core integrating between multi-dimensions, ranging from resources, reliability, to burden. This makes it possible to study whether adaptive control strategy is hopeful before introducing the full complexity of high-fidelity and multi-factor life-support simulation.

## Conclusions

This study developed a PPO-based regulation framework for micro closed-loop space station ecosystems and evaluated it under stochastic simulation with reliability decay and environmental coupling. Within this setting, the learned policy maintained resource safety margins and improved regulation smoothness and long-term ecological performance relative to threshold-based control. These results provide simulation-level evidence of control feasibility, while telemetry-informed validation and hardware-in-the-loop testing remain necessary for engineering deployment. Future work should focus more diverse fileds, ranging from telemetry-informed categroies, multi-factor robustness analysis to embedded deployment experiments. In that case, the proposed framework can be evaluated under more realistic operating conditions.



## References


1. Tipaldi, M., Iervolino, R., Massenio, P.R., “Reinforcement Learning in Spacecraft Control Applications: Advances, Prospects, and Challenges,” *Annual Reviews in Control* 54:1–23, 2022, doi:10.1016/j.arcontrol.2022.07.004.
2. Razzaghi, P., Tabrizian, A., Guo, W., Chen, S., et al., “A Survey on Reinforcement Learning in Aviation Applications,” *Engineering Applications of Artificial Intelligence* 136:108911, 2024, doi:10.1016/j.engappai.2024.108911.
3. Vega, L., “Environmental Control and Life Support (ECLS) Systems,” in Handbook of Bioastronautics, Springer, 2021, pp. 69–82, doi:10.1007/978-3-319-12191-8_132.
4. Gratius, N., Wang, Z., Hwang, M. Y., Hou, Y., et al., “Digital Twin Technologies for Autonomous Environmental Control and Life Support Systems,” *Journal of Aerospace Information Systems* 21(4):332–347, 2024, doi:10.2514/1.I011320.
5. Raihan, A., “Toward Sustainable Living in Space: A Review of Environmental Control and Life Support System Technologies,” *Space Habitation* 2(1):100047, 2026, doi:10.1016/j.spaceh.2025.100047.
6. Yang, K., Yang, C., Yang, H., Zhou, C., “Complexity Evaluation of an Environmental Control and Life-support System Based on Directed and Undirected Structural Entropy Methods,” *Entropy* 23(9):1173, 2021, doi:10.3390/e23091173.
7. Liu, H., Yao, Z., Fu, Y., Feng, J., “Review of Research into Bioregenerative Life Support System Which Can Support Humans Living in Space,” *Life Sciences in Space Research* 31:113–120, 2021, doi:10.1016/j.lssr.2021.09.003.
8. Ciurans, C., Bazmohammadi, N., Poughon, L., Vasquez, J. C., et al., “Hierarchically Controlled Ecological Life Support Systems,” *Computers & Chemical Engineering* 157:107625, 2022, doi:10.1016/j.compchemeng.2021.107625.
9. Porterfield, D. M., Tulodziecki, D., Wheeler, R., Cross, M. K. D., et al., “Critical Investments in Bioregenerative Life Support Systems for Bioastronautics and Sustainable Lunar Exploration,” *npj Microgravity* 11(1), 2025, doi:10.1038/s41526-025-00518-4.
10. Wilson, C., Riccardi, A., “Enabling Intelligent Onboard Guidance, Navigation, and Control Using Reinforcement Learning on Near-Term Flight Hardware,” *Acta Astronautica* 199:374–385, 2022, doi:10.1016/j.actaastro.2022.07.013.


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## Declaration of AI-Assisted Technologies

During the preparation of this manuscript, AI-assisted tools were used only for grammar checking and language refinement. All modeling, algorithm design, simulation development, result interpretation, and

scientific conclusions were independently completed and verified by the authors. The authors take full responsibility for the integrity and originality of the content.

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